

- **Introduction and Motivation.** Reasoning about action and event is one of the most mature research areas in Artificial Intelligence (AI): it is almost as old as AI itself. This research area has a direct connection to one of the central AI topics - namely, how to solve a problem using general domain independent techniques for finding a sequence of transformations or constructors/destructors that create a desired partially specified goal from a given initial state of affairs. This research area has strong foundations in mathematical logic. It is influenced and inspired by modern research in logic, natural language understanding, high-level robot control, database transactions, and several areas in computer science. The growth in this area was to a significant degree stimulated by research done in Toronto (Canada), and in a few other universities world-wide. This area consists of several sub-areas, where researchers are seeking different ways to overcome computational difficulties associated with reasoning about effects of actions (events) in large, incompletely known application domains that may include unspecified and unnamed objects. The areas of application are lifted planning in an open world with unknown objects, urban planning (including automated planning for transportation and parking), control of hybrid (mixed discrete-continuous) systems and embedded systems (also known as cyber-physical systems), problem solving, discovery in natural sciences, computing causes of failures from given traces (logs of events), integration of task and motion planning for robotics, verification of business processes, reasoning about streams of database transactions.

This course starts with basic introduction to mathematical logic: propositional logic and first order logic. The students are expected to learn a professional logical jargon and concepts that will help them to work on a higher level of abstraction. This logical language serves to provide a formal specification that facilitates a correct implementation in one of the programming languages. Moreover, using logical specification, it becomes possible to formulate interesting, general and realistic AI research problems that cannot be approached in other terms. After that, we review the most important results in this research area, and then focus on modern developments. We will discuss the compromises involved in providing useful logical representations that allow reasoning about actions to remain computationally tractable. There are some limits imposed by computational tractability; we will mention what we can do when trying to develop efficient AI implementations.

The assignments will usually include problem solving exercises. One of the assignments on planning may include also developing and running a benchmark problem using the state-of-the-art planners, using a provided deductive planner, and doing experimental

comparison of different planner configurations on several planning instances of the benchmark.

Topics may include: an algorithm for correct reasoning in propositional logic and in first order logic, the situation calculus, the projection problem (whether a given state can be reached after executing a given sequence of actions when initial states are not known completely), regression (reasoning backwards), progression (reasoning forward), the frame, and ramification problems, the precondition and successor state axioms, planning, execution monitoring, reasoning about effects of continuous and concurrent processes extended in time, identifying an actual cause of an observed effect from logs of events that happen during the run of a system.

It was long time recognized that actions have not only direct, but also indirect effects, and in many cases, have no effect on most of the features characterizing application domains. For example, pressing an electric switch button has a direct effect on the button - it becomes pressed - but also has an indirect effect, e.g., lights turn on. It turns out that chains of mutual dependencies leading to indirect effects may be quite complicated. We may consider representations developed to specify non-effects succinctly and to reason about indirect effects correctly. For the latter purpose, representations based on causality turned out to be most successful. We may explore some of the recent results about using causality to compile implicit indirect effects into explicit effects.

This course provides necessary and sufficient background for the students who would like to be involved in modern research on reasoning about actions in artificial intelligence.

- **Prerequisites** For undergraduate students: An undergraduate level course in AI (e.g., **CPS721** at TMU).

Since no previous knowledge of mathematical logic is expected, this course starts with a brief introduction into propositional logic, and first order logic. In addition, a popular reasoning algorithm based on linear resolution is also introduced and illustrated on several examples.

It is assumed that the students have the basic skills of programming in PROLOG, or they are willing to learn the basics of PROLOG by themselves. One of the homework assignments (on automated planning) will require understanding of a large PROLOG program and writing a short program in PROLOG. Several handouts include programs written in PROLOG.

- **Delivery Mode:** Lectures: 3 hours. Assigned readings. Out-of-class discussions in a lab.

The course is delivered in-person including 3 (2+1) hours of synchronous lectures per week (for 12 weeks) and 1h of synchronous lab/tutorial per week (for 10

weeks). Students are **not** allowed to record the lectures.

Some of the lectures may include problem solving sessions, where the instructor will solve problems similar to homework assignments and similar to exams. All the presentation slides, except of problem solving sessions, will be posted on D2L to facilitate asynchronous learning. The labs may include tutorials and solving typical problems similar to exams and homework assignments.

- **Calendar Description:** The course will focus on the theory and implementation of dynamical systems considered from the perspective of artificial intelligence. Modern logical representations of actions and their effects will be discussed in detail. The emphasis will be on the compromises required to ensure computational tractability of reasoning about effects of actions. The course will show how these research issues are relevant to artificial intelligence and to applications beyond the traditional area of artificial intelligence. Topics may include: logical foundations, algorithms for correct reasoning, automated heuristic lifted planning, causality, time, planning for mixed discrete-continuous systems, and reasoning about direct and indirect effects of actions.

- **References:**

There is **no** mandatory textbook. Handouts with slides will be posted on D2L after every class. In addition to attending classes and reading the handouts the students can read a few recommended books available from the TMU Library Reserve or as e-books.

The following textbooks are recommended (some of the readings are optional and/or supplementary to your lecture notes).

- The book [How to prove it : a structured approach](#) by Daniel J. Velleman provides foundations and a very readable introduction to a few basic topics. It can also serve well as a reference. Published by Cambridge University Press in 2006, 2nd edition, ISBN 978-0521675995 (paper-back). Alternatively, consider Edward A. Bender, S. Gill Williamson ``Short Course in Discrete Mathematics'', Dover, 2005. See sections on [Functions and Order](#) posted online in PDF. Alternatively, recall differences between *function* and *relation* from a short and readable overview of ``The Language and Grammar of Mathematics'', published as Section I.2 (Part 1 "Introduction"): pages 8-16, in Timothy Gowers (Editor) [The Princeton Companion to Mathematics](#), Princeton University Press, 2008.
- Uwe Schöningh ``[Logic for Computer Scientists](#)', published by Birkhäuser, Boston, 2008 ([Available online: see the TMU Library.](#))

- Mordechai Ben-Ari `` Mathematical Logic For Computer Science", corrected 6th printing of the 2nd edition, Springer 2003, or a more recent edition of this book. The recommended [3rd edition](#) of his book was published in 2012. [This is a textbook for a TMU course [MTH714](#).
- Stephen Cook: Slides for the course [CSC 438S/2404S](#): Computability and Logic. The following slides can be relevant: [Propositional Calculus](#), read pages 2-6 only, and [Predicate Calculus](#), read the pages 18-26 only.
- Elliot Mendelson `` Introduction to mathematical logic", any recent edition is fine.
- Joseph R. Shoenfield `` Mathematical Logic" (paperback). Publisher: A K Peters/CRC Press, 2001. [Chapters 2 and 3].
- Anil Nerode and Richard A. Shore `` Logic for applications", published by Springer, 2nd edition, 1997. ISBN: 978-0387948935. Available in the [university library](#).
- Raymond Reiter `` [Knowledge in Action](#): Logical Foundations for Specifying and Implementing Dynamical Systems". MIT Press, 2001. [[Available online from the TMU library](#): Chapters 3, 4, 5, 7, 9, 10.]
- Hector Geffner and Blai Bonet: [A Concise Introduction to Models and Methods for Automated Planning](#): Read Chapter 2. This book is available [online from the TMU Library](#). This book was published in June 2013, by Morgan and Claypool publishers, in the series Synthesis Lectures on Artificial Intelligence and Machine Learning (doi:10.2200/S00513ED1V01Y201306AIM022).
- S. Russell and P. Norvig "Artificial Intelligence", published by Pearson. The following chapters are relevant: [Planning \(Chapter 11\)](#) from the 2nd edition, 2003, p. 375-416, ISBN 0137903952 and [Solving Problems by Searching \(Chapter 3\)](#), from the 4th edition, p. 59-93, or from [3rd edition](#), pages 65-113. This textbook is [available from the university library](#).
- Frank van Harmelen, Vladimir Lifschitz, Bruce Porter (Editors) " [Handbook of Knowledge Representation](#)", published by ELSEVIER, 2007, 1034 Pages, ISBN: 978-0-444-52211-5. [Chapters 3, 16, 23.] Available in the university library.
- Optional/required readings may also include a selection of research papers published in journals or in the proceedings of international conferences.

- **Student Evaluation** See the table below with a brief description and a tentative calendar for all course work. To pass the course the following is required:
 - At least 50% must be achieved on the theoretical component (the weighted total of the quiz, test, and exam marks)
 - At least a 50% grade on the weighted total lab mark
 - At least a 50% grade on the remaining practical component (the weighted total of the assignment and project marks)

In addition, any student who missed half or more than half of the lectures fails the course for non-attendance. Class attendance is mandatory.

- **Lecture Topics (tentative list):**

1. Introduction to propositional logic. Syntax: well-formed formulas, sub-formulas of a formula, defined connectives. Semantics: truth assignment, satisfiability, tautology, logical consequence, well-known propositional equivalences. DNF (disjunctive normal form), CNF (conjunctive normal form), NNF (negation normal form). The satisfiability problem for a formula in propositional logic. The entailment problem. Reasoning in propositional logic using resolution and its refinements: set-of-support strategy, linear resolution, SLD-resolution. Horn formulas. Soundness and completeness of a resolution algorithm for solving the satisfiability problem.
2. Introduction to first order logic (FOL). Syntax: predicate symbols, function symbols, terms, quantifiers, free and bound variables. Semantics: interpretation (structure), object assignments, model, basic semantic definition (BSD), satisfiability, logical consequence, validity, examples of FOL equivalences.
3. How to prove logical consequence and logical equivalence using BSD: examples. Substitution in terms and in formulas, a term freely substitutable for a variable in a formula, substitution theorem. Prenex normal form (PNF). Skolemization. Substitution, unification of terms (the unification algorithm is optional), examples. Herbrand's Theorem. Horn formulas. Reasoning in FOL using resolution and unification. Examples of refutations using resolution. Godel's soundness and completeness theorem for FOL.
4. Examples of counter-models for some satisfiable FOL formulas that are not valid. The satisfiability and entailment problems in FOL are undecidable (Church-Turing theorem). Compactness of FOL. A theory of single successor, its axiomatization, and models, including non-standard models. Optional: finitely axiomatizable theories, incomplete and complete theories, e.g., linear dense order.

Limitations of FOL: transitive closure cannot be correctly formulated in FOL. Second order logic: a brief overview of syntax and semantics. DATALOG.

5. Introduction to reasoning about action. Intuitive ontology for the situation calculus. The qualification problem: the precondition axioms as a tentative solution in FOL.

Deterministic, primitive, atemporal actions without side-effects. Frame axioms (axioms about lack of effects for actions), the [frame problem](#): Reiter's solution. Effect axioms, normal form for effect axioms. Transforming effect axioms for a given fluent into a single positive effect axiom and a single negative effect axiom for the fluent. Explanation closure, causal completeness assumption, successor state axioms.

6. Foundational axioms for the situation calculus, the tree of situations. Principles of induction. Uniform formulas, regressable formulas. Basic action theories (BATs): precondition axioms, successor state axioms, initial theory, foundational axioms, unique name axioms (UNA). The projection and executability problems. Two techniques for solving these problems: regression (reasoning backwards) and progression (reasoning forward). The regression operator. The relative satisfiability theorem. The theorem about reducing the projection problem to entailment of a regressed formula from an initial theory (together with UNA).

7. Data base update transactions formalized in the situation calculus. [STRIPS assumption vs ADL theories vs arbitrary basic action theories](#). Implementing basic action theories in Prolog: closed world assumption (CWA), open world assumption (OWA). Domain Closure Assumption (DCA) vs open domains with possibly unspecified objects.

8. The planning problem formulated as an entailment problem in the situation calculus. Advantages of this formulation. Optimal planning: A* algorithm. Priority Queue data structure. Domain independent heuristics: MAX, ADD. Planning graph based heuristics: FF. State of the art planners: Fast Downward, Best-First Width Search. Deductive planning based on situations and fluents.

9. Actual causes in situation calculus. Motivation: token causality vs generic causality. Well-known examples of finding causes from a given scenario. Pearl and Halpern's structural causal models approach. Using basic action theory to represent domain dynamics. One-step regression for a formula representing an observed effect. Achievability cause, and the chain of achievability causes. Maintenance causes serve to mitigate risks associated with potential threats to an achieved effect. In a general case, actual causes recursively combine achievability and maintenance causes. Applications of causal analysis to computing the actual cause of a failure from a log of actions/events.

10. Reasoning forward in the situation calculus. Forgetting about a single ground atom and about multiple ground atoms. Progression in the situation calculus. Local-effect basic action theories. Computing progression efficiently.

11. Advanced topics in reasoning about action: time and continuous processes. Instantaneous actions, processes extended in time. Natural actions: representing physical laws, least natural time point. Planning in mixed discrete-continuous (hybrid) systems. Correct reasoning about hybrid systems. Diagnosis of failures in hybrid systems. Approaches to axiomatizing concurrency.

12. (Optional topic, if time permits) Reasoning about indirect effects of actions. Circumscription: general definition, circumscription with fixed and variable predicates, examples when circumscription can be expressed in FOL. State constraints, the ramification and qualification problems, the causality predicate, causal rules, compiling state constraints into successor state axioms.

13. (Optional topic, if time permits) Advanced topics in reasoning about action: stochastic actions. Markov Decision Processes (MDPs), decision theoretic (DT) planning. Decision-tree based methods to solving the DT-planning problem with a finite horizon. DT-Golog: problem representation, semantics, computing a policy and its value, sensing actions, on-line interpreter. Applications to robot programming and to requirements engineering.

14. (Optional topic, if time permits) Taxonomies of actions: troponyms (the possible relations between verbs in the semantic network of the WordNet database). Computational advantages of taxonomies of actions. Successor state axioms using taxonomies of actions.

Course Policies

- Grading requirements for undergraduate students: To pass the course the students have to get
 - At least 50% must be achieved on the theoretical component (the weighted total of the quiz, test, and exam marks)
 - At least a 50% grade on the weighted total lab mark
 - At least a 50% grade on the remaining practical component (the weighted total of the assignment and project marks)

See the table below with a brief description and a tentative calendar for all course work.

- For graduate students all course work mentioned in the table below is scaled down to 80%. Graduate students may be asked to do additional work on assignments and tests. The graduate students have to complete a small project that is worth of 20% of the final score. In addition, the graduate students may be asked to prepare slides for a 20-30min talk on a topic related to the course, and present their talk in class.
- The students are strongly encouraged to take notes in class, and study their notes after class. Learning can be a gradual process that requires time and efforts. The students benefit from attending lectures since some important details will be discussed only there. For this reason, attending lectures and (and extra tutorials/labs) is mandatory.
- ***All course materials posted on D2L and presented in class are copyrighted and protected by law. You cannot share them with anyone. You cannot repost them anywhere on the Web.*** Please review the [TMU policy](#) about copyrights. Moreover, you cannot share or repost any of your solutions to the labs or to assignments, since doing this would violate the cps822 policy on collaboration, and the TMU Academic Policy 60. You can read online [parts of this policy](#) related to "Academic misconduct".
- **Examinations:** The quizzes, a midterm exam, and the final exam may include problem solving, and short essay questions. The duration of these examinations may be around 15-20 min, 1h40min, and 2h30min, respectively. Extra quizzes (or polls) can be given at any time in class or during the labs without prior warning. The midterm and final exam may include questions similar to homework assignments and lab exercises. The final exam can be cumulative and may include all the material covered throughout the term with focus on the topics discussed after the midterm exam. There will be **no** supplemental examination. Grades are earned for the demonstration of knowledge.
- If you miss a midterm test, or a final exam for medical reasons, you have to submit an academic consideration request through the [Online Submission Form](#) and submit a copy of a completed [official Health Certificate](#) to the designated contact person or to the department of Computer Science within 3 working days. Once the submitted student's health documentation has been verified, the instructor will be notified of the verification. Similarly, all documentation related to special accommodation or [academic consideration](#) should be submitted online to the designated contact person within the specified time limits.

- Dates are subject to change, all changes will be announced in class and on the course Web pages.
- Assignments should be submitted on or before the deadline specified in the assignment (you are encouraged to submit assignments earlier). Your assignment is considered late if any part of the assignment is late (even if it is just 1 minute late) and the penalty for a late assignment is 10% off. No assignments will be accepted if more than one day late. Start solving your assignment on the same day when it is posted. Do not procrastinate. *No make-up assignments.*
- **Tutorials/Labs are mandatory.** Usually, each tutorial includes a mini-lecture and may include as well a poll/quiz. Each student must answer quiz questions individually. In addition, the TA who is teaching a lab can provide additional practice questions/quizzes. The students who actively participate in labs and volunteer during the lab to answer the questions of their T.A. may become eligible for extra 4% participation marks. Lab Marks: are given for attendance and answering correctly questions, as specified by T.A. The lab mark will be given only if
 1. the student attends the entire lab in the section where s/he is enrolled,
 2. the T.A. observed the student actively participating in discussion, solving questions, and
 3. the student signed up the solutions as required, provided a family name and a student number.
- 4. There will be **no** make-up labs. No video recordings of lab, or notes from the labs will be posted. The student who missed a lab should try to solve independently the exercises given during the lab time and verify the solutions with the T.A., or with peers in class.
 - From time to time, I will hand out exercises. The students are expected to solve the exercises, but they will not be graded. However, working on exercises will improve your understanding of this course and will help you to get better marks on tests.
 - Up to **4% extra credit** may be assigned for **active class participation**, e.g., a student attends the entire class, participates actively by asking/answering questions, works on problems in class and/or attends office hours. Class participation marks are given at discretion of the course instructor. They cannot be requested by the students.
 - Handouts and assignments will be made available on the Web only. You are responsible for visiting the course Web pages regularly and reading assignments related information that is provided or linked from these Web pages. Before sending

your questions by e-mail to the instructor, check these Web pages whether similar questions have been already answered.

- Email communication: please send your email from local TMU's email addresses only. Email sent from Bell, Rogers, Google and other popular providers can be filtered out as spam and might not reach the instructors. Email messages will be normally answered within 24 hours. However, messages sent on weekend (starting from Friday late afternoon) will be usually answered on Monday.
- Grades for labs, tests and assignments will be normally posted on my.torontomu.ca Web site no later than two weeks after the due date (exam date). Marking guides, the assignments and some other course related documents will be posted on my.torontomu.ca only. Feedback will be usually provided to students within two weeks. The students can contact the TA who was responsible for marking, if they have questions about marking, or attend the office hour. Missing marks for labs cannot be requested since only those students who attended a lab and actively participated are eligible to get a mark (see the policy for labs above).

Policy on collaboration in homework assignments

Discussing general approaches to problems is allowed. However, home work assignments are **individual**, unless group work is allowed in the assessment. No collaboration is allowed between groups when you write final solutions. You may discuss assessments only with one another student currently taking the course. However, you should **never put your name on anything you do not understand**. If challenged, *you must be able to reproduce and explain all solutions by yourself, or solve similar exercises*. If you cannot explain a solution that you handed in, or if you cannot solve an exercise similar to an assessment that you submitted, this will negatively affect your grade. In particular, you might be asked to solve exercises during the office hours, or in class (as a quiz). These unscheduled tests or evaluations can be given at any time without prior notice. Remember that if you work with a partner, you are still expected to know solutions of all exercises from the home work. Grades are earned for the demonstration of knowledge. In cases when a student fails to demonstrate knowledge about a home work, the grade for the home work can be decreased to 0. The first page of your homework should include: the name of all students with whom you discussed any homework problems (even briefly). Otherwise, it is assumed that you didn't discuss with anyone except the instructor. Copied work (both original and copies) will be graded as 0. An additional penalty for copied work may be assigned up to - 2% of the final course grade, i.e., cheating students can get a negative grade on an assignment. Additional penalty for a copied (in part or in whole) solution to a quiz can be up to -1% of the final course score. This is in accordance with the new TMU Senate Policy 60

on Academic Integrity. Repeated involvement with plagiarism will be penalized in accordance with the departmental policy and the Student Code of Academic Conduct.

Contract Cheating Statement

In regard to any and all assessments in this course, the use of Chegg or any other similar help site/service will be pursued as "*contract cheating*".

The use of ChatGPT, Gemini, Copilot and similar generative large language models (LLM) with the purposes of solving homework problems will be pursued as "*a breach of Policy 60: Academic Integrity*", if the student accessed them before submitting course work and assessment is presented as if it is one's own original work without appropriate referencing. Generative LLM tools may only be used for comparison with your own course work that you have already submitted, but not for the creation of submitted work.

You might be allowed to use generative LLM tools to complete your end of the term Project. All content automatically generated by LLM tool(s) should be properly cited and referenced in your project report using different fonts and/or colours to make sure your own writing is visibly distinct from any automatically generated content. An additional policy on how to complete your Project will be posted at D2L.

Falsified citations or misrepresentation of source material will be considered a breach of Policy 60. You are responsible for the accuracy of the work you submit.

In regard to any and all assessments in this course, the use of any third party (e.g., family member, freelancer, room-mate, friend, tutor) to complete work on your behalf will be pursued as "*contract cheating*" under Policy 60 "Academic Integrity".

Policy 60 Penalty Guidelines for contract cheating (e.g., viewing a solution on Chegg or Discord) that only impacts you: **F in course**.

Policy 60 Penalty Guidelines for contract cheating that facilitates cheating for others (e.g., posting a question to Chegg): **Disciplinary Suspension**.

ACADEMIC MISCONDUCT

Committing academic misconduct, such as plagiarism and cheating, will trigger academic penalties including failing grades, suspension and possibly expulsion from the University. As a TMU student, you are responsible for familiarizing yourself with the [Student Code of Academic Conduct](#).

Policy on Non-Academic Conduct

No disruption of instructional activities is allowed. In particular, taking video/photos in class is strictly prohibited since this violates the copyright and privacy policies. Among many other infractions, [the Code](#) specifically refers to the following as a violation:

` ` Disruption of Learning and Teaching - Students shall not behave in disruptive ways that obstruct the learning and teaching environment." In particular, the students can use the laptops (and similar electronic devices) in class only for taking notes.

The policy for remote synchronous content delivery over Zoom: the students are expected to use a Chat Window only for asking or answering lecture-related questions, but not for personal communication. The students can click on a "Raise a hand" icon in the Participants window in Zoom, if they have a questions, or unmute themselves and ask a question verbally. The students are expected to pay attention to a lecture and volunteer to answer instructor's questions during the class-time. The students are normally expected to keep their mics mute and unmute them only to answer instructor's questions. Make sure there are no background noises and no audio distractions in your environment.

In the case of in-person classes, in order to create an environment conducive to learning and respectful of others rights, phones and pagers must be silenced during lectures.

Students should refrain from arriving late and/or leaving the classroom before the lecture is finished. In difficult cases, penalties can be imposed by the Student Conduct Officer.